ETF Factor Rotation Project

A Python-Based Benchmark-Relative Factor Allocation Framework

From ETF factor investing theory to automated portfolio analytics reporting

By: Elias BANON

Academic Year: 2025–2026

Table of Contents

1. Executive Summary

This report presents the ETF Factor Rotation Project, a Python-based portfolio analytics framework focused on the U.S. equity market and built around investable factor ETFs. The analysis focuses on four exposures: Value, Momentum, Quality and Low Volatility. It evaluates the dynamic allocation strategy against three complementary reference portfolios. ITOT, the iShares Core S&P Total U.S. Stock Market ETF, serves as the broad-market benchmark. LRGF, the iShares U.S. Equity Factor ETF, provides an investable multifactor benchmark. The strategy is also compared with a simple equal-weight portfolio of the four selected factor ETFs making it possible to assess whether the dynamic allocation rules add value beyond passive factor diversification. The central question is therefore practical: can a transparent factor rotation strategy improve risk-adjusted performance relative to its benchmarks? Rather than relying on discretionary market views, the project addresses this question through explicit and reproducible allocation rules.

Each factor is scored against ITOT using Information Ratio, Sharpe advantage, volatility advantage and drawdown advantage. The variables are normalized cross-sectionally, converted into monthly weights and capped at 80% per factor. Any residual allocation is assigned to the benchmark. Over the comparable sample ending on 30 June 2026, the active factor selection portfolio delivered 16.69% annualized return, compared with 14.78% for ITOT. Its Sharpe ratio was 0.89 versus 0.79 for the benchmark and its maximum drawdown was -33.61% versus -35.00%.

The project's main contribution lies in its complete analytical chain. It begins with a practical investment question and translates systematic factor exposures into an investable ETF universe. The selected ETFs are then evaluated relative to ITOT and the resulting signals are converted into a monthly allocation rule with explicit controls against look-ahead bias. Finally, an automated factsheet transforms the model's outputs into professional portfolio reporting. Given the methodological limitations discussed in the report, the project should be interpreted primarily as a portfolio analytics framework rather than as a fully validated investment strategy.

2. Project Rationale

This project extends my dissertation research on how ETFs are reshaping asset management. The dissertation argues that ETFs are no longer simply low-cost investment products. They have become modular portfolio-building tools that are changing the economics, investment strategies, distribution models and market infrastructure of the industry. By making broad market exposure cheaper, more transparent, more liquid and more immediate to access, ETFs reduce the value associated with simple market access. Value creation therefore increasingly shifts toward portfolio architecture, allocation discipline, risk monitoring, execution quality and client reporting. This project translates that broader industry transformation into a concrete Python application using listed factor ETFs as investable building blocks within a rules-based allocation and reporting process.

Factor ETFs are a natural case study. They transform systematic investment styles into listed instruments. Value, Momentum, Quality and Low Volatility can then be observed through prices, compared with a benchmark and combined inside a portfolio. The objective is to build a transparent and reproducible portfolio analytics framework that translates factor signals into investable ETF allocations and professional monthly reporting.

3. ETF Factor Investing

This question matters in ETF-based portfolio construction. ETFs allow investors to separate broad market exposure from more specific return drivers. Factor ETFs fit this logic because they make systematic styles tradable and measurable:

- Value seeks companies that appear inexpensive relative to fundamentals.
- Momentum captures companies that have recently outperformed.
- Quality focuses on profitability, balance-sheet strength and earnings stability.
- Low Volatility targets companies with historically lower price fluctuations.

The key issue is cyclicality. Factors do not lead the market continuously and their relative performance changes across regimes. A static allocation assumes that the same mix remains suitable over time. A dynamic framework instead accepts that factor conditions can change. The model translates these changing benchmark-relative characteristics into monthly weights. The next step is therefore to define a clean ETF universe and a precise data protocol before building the allocation rule.

4. ETF Universe and Reproducibility Protocol

4.A ETF universe

To keep the question practical, the project uses real ETF prices rather than theoretical factor portfolios. This makes the analysis closer to implementation, although each ETF remains an imperfect proxy for its underlying factor.

Role	ETF Ticker	Purpose
Value	VLUE	Value factor proxy
Momentum	MTUM	Momentum factor proxy
Quality	QUAL	Quality factor proxy
Low Volatility	USMV	Low-volatility factor proxy
Benchmark	ITOT	Broad U.S. equity benchmark
Multifactor comparator	LRGF	Multifactor ETF comparator

The universe is consistent. It relies on the iShares ecosystem. Consequently, this structure keeps the comparison focused on factor allocation rather than provider differences.

4.B Reproducibility protocol

The following protocol makes the model reproducible without forcing the reader to infer technical settings from the notebook.

Item	Implementation choice
Price source	Yahoo Finance using yfinance
Price field	Adjusted prices using auto_adjust=True

Dividend and split treatment	Dividends and splits are incorporated through adjusted prices
Start date	1 January 2016
Comparable sample in factsheet	1 February 2017 to 30 June 2026
Return frequency	Daily percentage returns
Missing data treatment	Dates with missing prices for any ETF are removed
Trading-day convention	252 trading days per year
Rolling window	252 trading days
Risk-free rate	0%
Benchmark	ITOT
Rebalancing frequency	Monthly
Signal date	Last available trading observation at month-end
Weight application	Weights are forward-filled daily and shifted by one trading day
First effective date	Next available trading day after the signal date
Maximum factor cap	80% per factor
Residual allocation	Assigned to ITOT

Adjusted prices are used to capture each ETF's total return including dividends and stock splits. Missing observations are then removed to ensure that all portfolios are evaluated over the same trading dates. This creates a consistent basis for performance comparison. The Sharpe ratio is calculated using a 0% risk-free rate as a simplifying assumption. Since the same convention is applied to every strategy, relative Sharpe comparisons remain informative. However, the absolute Sharpe ratios should be interpreted with caution because short-term interest rates changed materially over the sample period.

5. Methodology

5.A Performance metrics

Once the data are aligned, portfolio performance is assessed using standard return and risk indicators. The first is annualized return which is calculated from compounded daily returns rather than from a simple arithmetic average.

Metric	**Formula / convention**
Daily return	$rd = Pd / P(d\text{-}1) - 1$
Annualized return	$((1 + rd)^{(252 / N)}) - 1$

Annualized volatility	sample standard deviation of daily returns x sqrt(252)
Sharpe ratio	(annualized return - 0%) / annualized volatility
Maximum drawdown	minimum value of cumulative performance / running peak - 1
Calmar ratio	annualized return / absolute maximum drawdown
Information Ratio	annualized active return / annualized tracking error

5.B Active score construction

Each factor is then evaluated against ITOT through four dimensions: Information Ratio, Sharpe advantage, volatility advantage and drawdown advantage. The score is simple by design but each component is made explicit.

Component	**Definition**
Information Ratio	Annualized mean of daily active returns / annualized tracking error
Sharpe advantage	Factor Sharpe ratio -ITOT Sharpe ratio
Volatility advantage	ITOT annualized volatility - factor annualized volatility
Drawdown advantage	Factor drawdown - ITOT drawdown

At each rebalancing date, every indicator is standardised across the four factors. For each factor, the z-score measures how far its value is above or below the cross-sectional average expressed in units of sample standard deviation: $z_{i,t}$ = ($x_{i,t}$ - mean(x_t)) / sample_std(x_t). The monthly active score is then defined as: Active_Score$_{i,t}$ = 0.40 * Z(IR$_{i,t}$) + 0.30 * Z(Sharpe_Advantage$_{i,t}$) + 0.15 * Z(Volatility_Advantage$_{i,t}$) + 0.15 * Z(Drawdown_Advantage$_{i,t}$).

5.C Allocation rule and cap treatment

The allocation process begins with factor selection. A factor is eligible only if both its Information Ratio and Active Score are positive. This first filter ensures that negative signals receive no allocation. Once the eligible factors have been identified, their positive scores are converted into preliminary portfolio weights. These weights determine the relative importance of each selected factor. However, to limit excessive concentration, the allocation to any single factor is capped at 80%. This cap directly determines the role of the benchmark in the portfolio. If a factor's preliminary weight exceeds 80%, the excess is not redistributed to the other factor ETFs. Instead, it is allocated to ITOT. More generally, any portfolio weight left after factor selection and capping is assigned to the benchmark. As a result, ITOT acts as the portfolio's residual exposure. This prevents the model from forcing a full allocation to factors when only one or two signals are attractive. It also explains why a concentrated factor position can coexist with a significant broad-market allocation. Finally, the strategy remains long-only throughout the process. Factor weights cannot be negative, and the combined allocation to the selected factor ETFs and ITOT always equals 100%.

5.D Timing convention and look-ahead control

Monthly signals are sampled at the last available trading observation of each month.The corresponding portfolio weights are then applied from the next trading day onward. This one-day delay ensures that month-end information is not used to explain the return observed on the same day. It therefore reduces look-ahead bias and makes the backtest more reliable.

5.E Complementarity and redundancy of score components

The score combines related indicators which creates a risk of partial redundancy. This is most relevant for Information Ratio, Sharpe advantage and volatility advantage. The design is therefore not presented as optimal. The components are still informative because they capture different angles. Information Ratio focuses on active return per unit of tracking error. Sharpe advantage compares total risk-adjusted performance. Volatility advantage isolates absolute risk reduction. Drawdown advantage adds a downside-risk dimension that volatility alone may miss. Consequently, a more advanced version should test whether removing one component materially changes allocations and results. In this report, the score is treated as a transparent design choice rather than a statistically optimized formula.

6. Results and Interpretation

6.A Main performance comparison

With these conventions in place, the performance comparison can be interpreted consistently. Over the comparable sample ending on 30 June 2026, the active portfolio outperformed the ITOT. The result came mainly from higher annualized returns. Volatility was almost identical to the benchmark, while the maximum drawdown was slightly less severe.

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar
ITOT	14.78%	18.67%	0.79	-35.00%	0.42
LRGF	13.39%	18.69%	0.72	-36.03%	0.37
Equal-Weight	14.73%	17.68%	0.83	-34.98%	0.42
Active	16.69%	18.69%	0.89	-33.61%	0.50

Relative to ITOT, the active portfolio generated +1.91 percentage points of annualized return and a Sharpe improvement of +0.10. The maximum drawdown improvement was +1.39 percentage points and the volatility difference was close to zero. The result is therefore best interpreted as return-driven outperformance rather than a major reduction in risk.

6.B Reading the allocation and cyclicality charts

The allocation chart adds context to this result. It shows that the strategy can become concentrated when one factor dominates the score. The recent 80% Value allocation illustrates this feature. It is consistent with the stated cap but it also increases dependence on one factor at a specific point in time. The cyclicality chart explains why the framework is useful. Factor leadership changes across the sample which supports the need for monitoring. However, the chart should not be read as a factor-by-factor attribution of portfolio performance. The charts therefore clarify the allocation process but they do not identify the precise source of the +1.91 percentage-point annualized return gap.

6.C Interpreting the +1.91 percentage-point gap

The +1.91 percentage-point annualized return advantage is an aggregate portfolio outcome. The available evidence indicates that the improvement was mainly return-driven rather than volatility-driven. The active portfolio earned more while taking a level of volatility close to ITOT. The report does not claim a full performance attribution. It does not decompose excess return by factor, by year, by monthly hit rate or by contribution from the residual benchmark sleeve. Those calculations would be needed to state precisely which exposures and periods produced the total gap.

This limits the strength of the conclusion. The result supports the usefulness of the framework as a transparent allocation and reporting process. It does not identify a stable alpha source or attribute the whole performance gap to Value, Momentum, Quality, Low Volatility or the benchmark sleeve.

6.D Automated monthly factsheet

The final layer of the project is the automated monthly factsheet. It reports key data, ETF universe, cumulative performance, annualized metrics, trailing returns, month-end allocation, strategy comparison, factor cyclicality and methodological limits. This reporting layer turns the model into a communication tool. In asset management, an analytical framework becomes more valuable when its assumptions, results and risks can be explained quickly and consistently.

7. Limitations and Critical Reflection

7.A Robustness as a limit, not a claimed result

This version does not include a full robustness analysis. It presents a transparent baseline framework rather than a statistically validated strategy. A complete validation would test alternative rolling windows, caps, score weights, cost assumptions and out-of-sample periods. These extensions could change the results. So, they are identified as future work rather than treated as completed evidence.

7.B Implementation realism and turnover

The project does not calculate turnover, the average number of allocation changes or the average holding period of each factor. It also does not apply a transaction-cost schedule. This limits the assessment of implementation realism. The reported +1.91 percentage points of annualized return should therefore be read before full trading frictions. Monthly rebalancing is more realistic than daily trading. However, bid-ask spreads, taxes, turnover and market-impact assumptions are still absent. A future version should report annual turnover and apply simple cost assumptions. This would test whether the excess return remains economically plausible after implementation costs.

7.C Risk-free rate convention

The 0% risk-free rate simplifies the Sharpe calculation. It is acceptable for a first implementation because every strategy is compared under the same convention. The convention remains imperfect. From 2017 to 2026, short-term rates changed materially. A time-varying risk-free rate would improve the absolute interpretation of Sharpe ratios, even if the relative comparison remains informative.

7.D Concentration, timing and ETF proxy limits

The 80% cap allows the model to express strong factor signals. It also creates concentration risk. A lower cap would reduce this risk but might also reduce the ability to capture factor leadership. The cap is therefore a portfolio construction assumption not a neutral technical detail. Timing is another limit. The model reacts after signals are observed. So, short factor rotations may be partly missed.

Finally, ETF proxies are not pure factors. VLUE, MTUM, QUAL and USMV have index rules, sector exposures, rebalancing policies and liquidity profiles. Their returns reflect both the intended factor and the specific ETF methodology.

8. Contribution

The project links finance reasoning, coding and professional communication. It starts from an asset-management question and converts it into data, methodology, portfolio analytics and reporting.

1. **Financial reasoning:** the project connects ETFs, factor investing, benchmark comparison and risk-adjusted performance.
2. **Python implementation:** the notebook covers price retrieval, return calculation, rolling metrics, scoring, allocation and reporting.
3. **Portfolio analytics:** the framework compares strategies through return, volatility, drawdown, correlation, Sharpe ratio and Calmar ratio.
4. **Critical thinking:** the report identifies limits around robustness, turnover, attribution, score design and concentration.
5. **Professional reporting:** the automated factsheet turns quantitative outputs into an asset-management-style deliverable.

The project also supports my broader interest in ETF-based asset management. It shows that ETFs can be analyzed not only as access products but also as instruments for exposure design, allocation monitoring and professional communication.

9. Conclusion

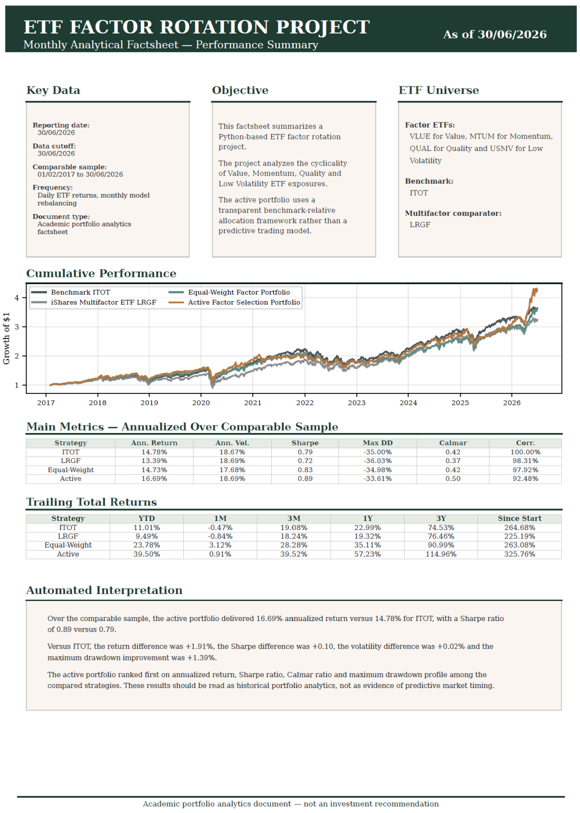
The ETF Factor Rotation Project shows how investable factor ETFs can be used inside a transparent benchmark-relative allocation framework focused on the U.S. equity market. It studies Value, Momentum, Quality and Low Volatility exposures. Then, it compares a dynamic portfolio with ITOT, LRGF and a static equal-weight factor allocation. The active portfolio improved annualized return, Sharpe ratio and drawdown profile over the comparable sample ending on 30 June 2026. This result is useful but it remains historical, parameter-dependent and gross of full implementation costs.

The main contribution is the process itself: a clear finance question, investable ETF proxies, a reproducible method, look-ahead control and an automated monthly factsheet. This makes the project relevant because it connects asset-management reasoning, Python implementation and professional reporting.

10. Appendix

10.A Monthly Analytical Factsheet

The following factsheet is generated from the Python notebook and uses the same reporting date, data cut-off and performance conventions as the report. The figures in the body of the report are aligned with this appendix.



ETF FACTOR ROTATION PROJECT
Monthly Analytical Factsheet — Performance Summary
As of 30/06/2026

Key Data

Reporting date: 30/06/2026
Data cutoff: 30/06/2026
Comparable sample: 01/02/2017 to 30/06/2026
Frequency: Daily ETF returns, monthly model rebalancing
Document type: Academic portfolio analytics factsheet

Objective

This factsheet summarizes a Python-based ETF factor rotation project.

The project analyzes the cyclicality of Value, Momentum, Quality and Low Volatility ETF exposures.

The active portfolio uses a transparent benchmark-relative allocation framework rather than a predictive trading model.

ETF Universe

Factor ETFs: VLUE for Value, MTUM for Momentum, QUAL for Quality and USMV for Low Volatility
Benchmark: ITOT
Multifactor comparator: LRGF

Cumulative Performance

Main Metrics — Annualized Over Comparable Sample

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Corr.
ITOT	14.78%	18.67%	0.79	-35.00%	0.42	100.00%
LRGF	13.39%	18.69%	0.72	-36.03%	0.37	98.31%
Equal-Weight	14.73%	17.68%	0.83	-34.98%	0.42	97.92%
Active	16.69%	18.69%	0.89	-33.61%	0.50	92.48%

Trailing Total Returns

Strategy	YTD	1M	3M	1Y	3Y	Since Start
ITOT	11.01%	-0.47%	19.08%	22.99%	74.53%	264.68%
LRGF	9.49%	-0.84%	18.24%	19.32%	76.46%	225.19%
Equal-Weight	23.78%	3.12%	28.28%	35.11%	90.99%	263.08%
Active	39.50%	0.91%	39.52%	57.23%	114.96%	325.76%

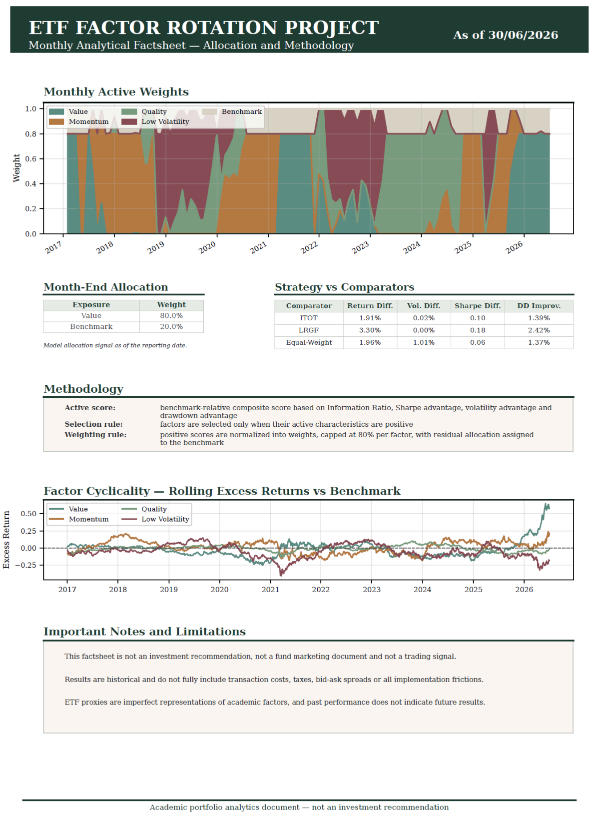
Automated Interpretation

Over the comparable sample, the active portfolio delivered 16.69% annualized return versus 14.78% for ITOT, with a Sharpe ratio of 0.89 versus 0.79.

Versus ITOT, the return difference was +1.91%, the Sharpe difference was +0.10, the volatility difference was +0.02% and the maximum drawdown improvement was +1.39%.

The active portfolio ranked first on annualized return, Sharpe ratio, Calmar ratio and maximum drawdown profile among the compared strategies. These results should be read as historical portfolio analytics, not as evidence of predictive market timing.

Academic portfolio analytics document — not an investment recommendation



ETF FACTOR ROTATION PROJECT
Monthly Analytical Factsheet — Allocation and Methodology
As of 30/06/2026

Monthly Active Weights

Month-End Allocation

Exposure	Weight
Value	80.0%
Benchmark	20.0%

Model allocation signal as of the reporting date.

Strategy vs Comparators

Comparator	Return Diff.	Vol. Diff.	Sharpe Diff.	DD Improv.
ITOT	1.91%	0.02%	0.10	1.39%
LRGF	3.30%	0.00%	0.18	2.42%
Equal-Weight	1.96%	1.01%	0.06	1.37%

Methodology

Active score: benchmark-relative composite score based on Information Ratio, Sharpe advantage, volatility advantage and drawdown advantage
Selection rule: factors are selected only when their active characteristics are positive
Weighting rule: positive scores are normalized into weights, capped at 80% per factor, with residual allocation assigned to the benchmark

Factor Cyclicality — Rolling Excess Returns vs Benchmark

Important Notes and Limitations

This factsheet is not an investment recommendation, not a fund marketing document and not a trading signal.

Results are historical and do not fully include transaction costs, taxes, bid-ask spreads or all implementation frictions.

ETF proxies are imperfect representations of academic factors, and past performance does not indicate future results.

Academic portfolio analytics document — not an investment recommendation

10.B Structure

The supporting notebook contains the complete analytical workflow, from ETF universe definition and price retrieval to data cleaning, return calculation, rolling analysis, drawdown measurement, active scoring, monthly allocation and automated factsheet generation. The project is organised around the following deliverables:

- **notebook.ipynb**: main Python notebook containing the full analytical workflow, calculations and explanatory comments.
- **README.md**: short project overview presenting the objective, data, methodology and main outputs.
- **outputs/**: Folder containing the charts, tables and factsheets generated by the notebook.
- **report.pdf**: Professional report designed to present the project in academic applications or interviews.
- **factsheet.pdf**: Automated monthly analytical factsheet generated directly from the notebook.